Artificial Intelligence Methods and Artificial Intelligence-Enabled Metrics for Surgical Education: A Multidisciplinary Consensus

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Abstract

Artificial intelligence (AI) methods and AI-enabled metrics hold tremendous potential to advance surgical education. Our objective was to generate consensus guidance on specific needs for AI methods and AI-enabled metrics for surgical education. The study included a systematic literature search, a virtual conference, and a 3-round Delphi survey of 40 representative multidisciplinary stakeholders with domain expertise selected through purposeful sampling. The accelerated Delphi process was completed within 10 days. The survey covered overall utility, anticipated future (10-year time horizon), and applications for surgical training, assessment, and feedback. Consensus was agreement among 80% or more respondents. We coded survey questions into 11 themes and descriptively analyzed the responses. The respondents included surgeons (40%), engineers (15%), affiliates of industry (27.5%), professional societies (7.5%), regulatory agencies (7.5%), and a lawyer (2.5%). The survey included 155 questions; consensus was achieved on 136 (87.7%). The panel listed

6 deliverables each for AI-enhanced learning curve analytics and surgical skill assessment. For feedback, the panel identified 10 priority deliverables spanning 2-year (n = 2), 5-year (n = 4), and 10-year (n = 4) timeframes. Within 2 years, the panel expects development of methods to recognize anatomy in images of the surgical field and to provide surgeons with performance feedback immediately after an operation. The panel also identified 5 essential that should be included in operative performance reports for surgeons. The Delphi panel consensus provides a specific, bold, and forward-looking roadmap for AI methods and AI-enabled metrics for surgical education.